Utils V8 — Library for Coolmay FX3G PLC

Utils is a library of general-purpose functions and function blocks for Mitsubishi FX3G PLCs programmed in GX Works 2. It provides bit-level WORD/DWORD manipulation, linear and non-linear scaling, sign negation, on/off hysteresis regulators, index rotation and shifting, progress calculation, a 3-position valve controller, analog-input scaling for the host PLC and L02 modules, and L02 IP configuration.

The functions are stateless: they return their result directly and are designed to be used inside expressions. The function blocks hold internal state and are called every scan.

Prerequisites

1. `FB_VALVE_3P` requires the **TimeControl** library: it reads the `TCO_DINT_50` 50 ms ticker, so the TimeControl ticker program must be installed and run in an interrupt task.

Terminology

The manual type names map to GX Works 2 types as follows.

Type in manual	Type in GX Works 2
`INT`	Word [Signed]
`WORD`	Word [Unsigned] / Bit String [16-bit]
`DINT`	Double Word [Signed]
`DWORD`	Double Word [Unsigned] / Bit String [32-bit]
`BOOL`	Bit

Architectural Description

The library is split into two groups of POUs:

- **Functions (`F_`)** — pure, stateless helpers that compute a value from their inputs and return it through the function name. They never access global data and may be nested inside expressions (`IF F_ISBON(D100, 2) THEN … END_IF`).
- **Function blocks (`FB_`)** — stateful blocks that hold internal state across scans (`FB_HYST`, `FB_VALVE_3P`, `FB_SCALE_AI`). They are declared as instances and called every scan.

Bit-manipulation functions operate on either a 16-bit `WORD` (`F_ISBON`, `F_SETB`, `F_RSTB`, `F_SRB`) or a 32-bit `DWORD` (`F_DISBON`, `F_DSETB`, `F_DRSTB`, `F_DSRB`). They take the device/value to modify and a 0-based bit number, and return the result directly.

Scaling is provided in three forms: linear 16-bit (`F_SCALE`), linear 32-bit (`F_DSCALE`), and non-linear through a point table (`F_SCALE_NL`). Both linear forms accept an `xNoLimit` flag that selects between clamping

(normalisation) and unclamped linear extrapolation. Analog inputs are scaled by the dedicated blocks `FB_SCALE_AI` (host PLC, `D8030` area) and `L02_SCALE_AI` (L02 modules, `R23700` area).

Function Blocks and Functions

Name	Type	Description
`F_ISBON`	Function	Test a single bit in a WORD
`F_DISBON`	Function	Test a single bit in a DWORD
`F_SETB`	Function	Set a single bit in a WORD and return the modified WORD
`F_DSETB`	Function	Set a single bit in a DWORD and return the modified DWORD
`F_RSTB`	Function	Reset a single bit in a WORD and return the modified WORD
`F_DRSTB`	Function	Reset a single bit in a DWORD and return the modified DWORD
`F_SRB`	Function	Set or reset a single bit in a WORD by a BOOL
`F_DSRB`	Function	Set or reset a single bit in a DWORD by a BOOL
`F_LIMIT`	Function	Clamp an INT value to a range
`F_SCALE`	Function	Linearly scale an INT between two ranges, optionally without clamping
`F_DSCALE`	Function	Linearly scale a DINT between two ranges, optionally without clamping
`F_SCALE_NL`	Function	Non-linearly scale an INT through a point table
`F_INCN`	Function	Increment an index with wrap-around
`F_SHFT`	Function	Shift an index within a range
`F_NEG`	Function	Negate an INT value
`F_DNEG`	Function	Negate a DINT value
`F_WORK_LEFT`	Function	Progress 100..0 from elapsed/total (INT units)
`F_WORK_LEFT_TIME`	Function	Progress 100..0 from elapsed/total (TIME)
`F_VALVE_POS`	Function	Valve position 0..1000 from elapsed/total TIME
`F_FLT10`	Function	Divide an INT by 10, returning REAL
`L02_SET_IP`	Function	Write L02 IP settings to R registers and apply them
`F_MBMOV`	Function	Block-move N words between devices
`FB_HYST`	Function Block	Heating on/off regulator with hysteresis

Name	Type	Description
FB_HYST_COOL	Function Block	Cooling on/off regulator with hysteresis
FB_SCALE_AI	Function Block	Scale a host-PLC analog input
L02_SCALE_AI	Function Block	Scale an L02-module analog input
FB_VALVE_3P	Function Block	3-position valve control with position feedback

Bit Functions

F_ISBON

Tests whether a given bit in a WORD is ON. The built-in BON instruction stores its result in a passed parameter instead of returning it; F_ISBON returns the result directly so it can be used inside expressions.

Variable	Scope	Type	Description
wIn	INPUT	WORD	The WORD to check
iBN	INPUT	INT	Bit number, 0-based, range 0..15

Example:

```
IF F_ISBON(D100, 2) THEN
    (* The third bit in D100 is ON *)
END_IF;
```

F_DISBON

DWORD version of F_ISBON.

Variable	Scope	Type	Description
dwIn	INPUT	DWORD	The DWORD to check
iBN	INPUT	INT	Bit number, 0-based, range 0..31

Example:

```
IF F_DISBON(D100, 31) THEN
    (* The highest bit of D100 is ON *)
END_IF;
```

F_SETB

Sets a single bit in a `WORD` and returns the modified WORD. The input is not changed.

Variable	Scope	Type	Description
`wIn`	INPUT	WORD	WORD to modify
`iBN`	INPUT	INT	Bit number to set, 0..15

Example:

```
wStatus := F_SETB(wStatus, 3);
(* Bit 3 of wStatus is now ON *)
```

F_DSETB

DWORD version of `F_SETB`.

Variable	Scope	Type	Description
`dwIn`	INPUT	DWORD	DWORD to modify
`iBN`	INPUT	INT	Bit number to set, 0..31

Example:

```
dwStatus := F_DSETB(dwStatus, 31);
```

F_RSTB

Resets a single bit in a `WORD` and returns the modified WORD. The input is not changed.

Variable	Scope	Type	Description
`wIn`	INPUT	WORD	WORD to modify
`iBN`	INPUT	INT	Bit number to reset, 0..15

Example:

```
wStatus := F_RSTB(wStatus, 3);
(* Bit 3 of wStatus is now OFF *)
```

F_DRSTB

DWORD version of `F_RSTB`.

Variable	Scope	Type	Description
`dwIn`	INPUT	DWORD	DWORD to modify

Variable	Scope	Type	Description
iBN	INPUT	INT	Bit number to reset, 0..31

Example:

```
dwStatus := F_DRSTB(dwStatus, 31);
```

F_SRB

Sets or resets a single bit in a WORD depending on xState and returns the modified WORD.

Variable	Scope	Type	Description
wIn	INPUT	WORD	WORD to modify
iBN	INPUT	INT	Bit number, 0..15
xState	INPUT	BOOL	TRUE = set the bit, FALSE = reset the bit

Example:

```
wStatus := F_SRB(wStatus, 15, TRUE);  (* set bit 15 *)
wStatus := F_SRB(wStatus, 0, FALSE);  (* reset bit 0 *)
```

F_DSRB

DWORD version of F_SRB.

Variable	Scope	Type	Description
dwIn	INPUT	DWORD	DWORD to modify
iBN	INPUT	INT	Bit number, 0..31
xState	INPUT	BOOL	TRUE = set the bit, FALSE = reset the bit

Example:

```
dwStatus := F_DSRB(dwStatus, 31, TRUE);
```

Value and Index Functions

F_LIMIT

Clamps iIn to the range [iMin..iMax] and returns the clamped value.

Variable	Scope	Type	Description
iMax	INPUT	INT	Upper limit
iIn	INPUT	INT	Value to clamp
iMin	INPUT	INT	Lower limit

Example:

```
iValue := F_LIMIT(100, iValue, 0);
(* iValue is now within 0..100 *)
```

F_SCALE

Linearly scales iVal from the input range iInLow..iInHigh to the output range iOutLow..iOutHigh and returns the result as an INT. Both ranges may be ascending or descending. The calculated value is converted to INT by truncation, not by rounding, so a result of 78.94 is returned as 78.

Clamping is controlled by xNoLimit:

- xNoLimit = FALSE — iVal is clamped to the input range before scaling, and the result is clamped to the output range. The returned value never leaves iOutLow..iOutHigh.
- xNoLimit = TRUE — no clamping is performed at all. Values outside the input range are extrapolated linearly, therefore the result may lie outside iOutLow..iOutHigh, including negative values.

Variable	Scope	Type	Description
iVal	INPUT	INT	Value to scale
iInLow	INPUT	INT	Input range minimum
iInHigh	INPUT	INT	Input range maximum
iOutLow	INPUT	INT	Output range minimum
iOutHigh	INPUT	INT	Output range maximum
xNoLimit	INPUT	BOOL	FALSE — clamp the input and the result to their ranges; TRUE — no clamping, extrapolation allowed

Example 1 — clamping (normalisation):

```
D8052 := F_SCALE(iPidTask, 0, 100, 0, 4000, FALSE);
```

iPidTask holds 0..100 and D8052 is the system register that controls analog output DA2, which accepts 0..4000. A task value of 80 produces 3200; a task value of 120 produces 4000, because the input is clamped to the input range before scaling.

Example 2 — extrapolation:

```
(* Display the negative deviation of a temperature sensor 0..100 -> 0..1000 *)
iDeviation := F_SCALE(iTemperature, 0, 100, 0, 1000, TRUE);
```

With `xNoLimit = TRUE` a temperature of `-5` produces `-50` instead of `0`, which is convenient when the output is a signed deviation rather than a physical actuator signal.

Example 3 — descending ranges:

```
(* Invert a 4..20 mA channel: raw 0..4092 -> 100..0 % *)
iDuty := F_SCALE(iRawAI, 0, 4092, 100, 0, FALSE);
```

A raw value of `0` produces `100`, a raw value of `4092` produces `0`. Either range, or both, may be given in descending order.

> **Note:** When `iInHigh` equals `iInLow` the input range has no slope. The function does not attempt the division and returns `iOutLow`.

> **Note:** Arithmetic is performed in `REAL` and converted back by truncation. A result computed by an exact rational formula may differ by **1** count, which is normally negligible for engineering units.

> **Note:** With `xNoLimit = TRUE` the returned value is not constrained to the `INT` range. An extrapolated result outside `-32768..32767` overflows the conversion and raises a PLC operation error, therefore the caller is responsible for keeping `iVal` inside a range that maps into `INT` limits.

F_DSCALE

`DINT` version of `F_SCALE`: linearly scales `diVal` from the input range `diInLow..diInHigh` to the output range `diOutLow..diOutHigh` and returns the result as a `DINT`. Both ranges may be ascending or descending. The calculated value is converted to `DINT` by truncation, not by rounding.

Clamping is controlled by `xNoLimit` in the same way as in `F_SCALE`:

- `xNoLimit = FALSE` — `diVal` is clamped to the input range before scaling, and the result is clamped to the output range. The returned value never leaves `diOutLow..diOutHigh`.
- `xNoLimit = TRUE` — no clamping is performed at all. Values outside the input range are extrapolated linearly, therefore the result may lie outside `diOutLow..diOutHigh`, including negative values.

Variable	Scope	Type	Description
`diVal`	INPUT	DINT	Value to scale
`diInLow`	INPUT	DINT	Input range minimum
`diInHigh`	INPUT	DINT	Input range maximum
`diOutLow`	INPUT	DINT	Output range minimum
`diOutHigh`	INPUT	DINT	Output range maximum

Variable	Scope	Type	Description
xNoLimit	INPUT	BOOL	FALSE — clamp the input and the result to their ranges; TRUE — no clamping, extrapolation allowed

Example 1 — clamping:

```
diResult := F_DSCALE(diVacuumSensor, 0, 32000, 0, 100000, FALSE);
```

This scales a vacuum sensor from 0..32000 raw units to 0..100000 Pa. A raw value of 40000 produces 100000, because the input is clamped before scaling.

Example 2 — extrapolation:

```
diDeviation := F_DSCALE(diPosition, 0, 100000, 0, 1000000, TRUE);
```

With xNoLimit = TRUE a position of -5000 produces -50000 instead of 0.

Example 3 — descending ranges:

```
diDuty := F_DSCALE(diTime, 0, 600000, 1000, 0, FALSE);
```

A time of 0 produces 1000, a time of 600000 produces 0. Either range, or both, may be given in descending order.

> **Note:** When diInHigh equals diInLow the input range has no slope. The function does not attempt the division and returns diOutLow.

> **Note:** The arithmetic is performed in REAL, which carries a **24-bit** mantissa. Values with more than 7 significant digits are rounded before the calculation, therefore large DINT operands (above 16777216) lose their low-order digits and the result may deviate from an exact rational computation. Time and sensor values in the usual PLC ranges are not affected.

> **Note:** With xNoLimit = TRUE the returned value is not constrained to the DINT range. An extrapolated result outside -2147483648..2147483647 overflows the conversion, therefore the caller is responsible for keeping diVal inside a range that maps into DINT limits.

F_SCALE_NL

Non-linearly scales iPV through a point table stored in consecutive D registers. The table starts at iDStart: the first register holds the number of points iPN, the next iPN registers hold the X coordinates, and the following iPN registers hold the Y coordinates.

Variable	Scope	Type	Description
iPN	INPUT	INT	Number of points
iDStart	INPUT	INT	First device of the point table

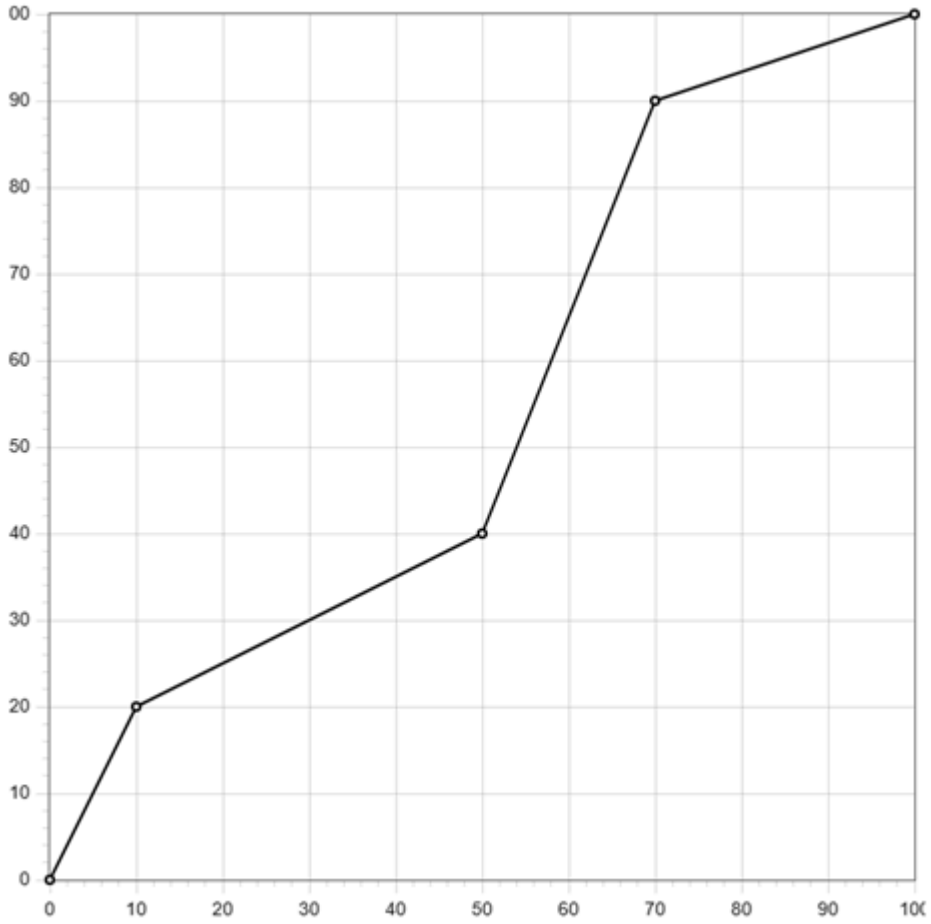
Variable	Scope	Type	Description
iPV	INPUT	INT	Process value on the X scale to convert to Y

For a 5-point table starting at D100:

```
D100 = 5      (number of points)
D101..D105 = X1..X5
D106..D110 = Y1..Y5
```

Example:

```
D200 := F_SCALE_NL(5, 100, iPv);
```



> **Note:** F_SCALE_NL writes the point count to D(iDStart) on every call and reads the X/Y coordinates through index registers. When a Coolmay panel is used, an XY Graph element can be pointed at the same D100 register to draw the curve.

F_INCN

Increments an index iCur by 1 when xIn is TRUE, wrapping to 0 after iMax. The result is returned.

Variable	Scope	Type	Description
xIn	INPUT	BOOL	Increment trigger
iCur	INPUT	INT	Current index
iMax	INPUT	INT	Maximum index (wrap point)

Example:

```
iCount := F_INCN(TRUE, iCount, 2);
```

iCount rotates 0, 1, 2, 0, 1, 2, … each scan.

F_SHFT

Shifts an index iIdx by iShifter within the range 0..iMax, wrapping around. It is used to rotate the starting index of an array (e.g. a cascade of pumps or heaters).

Variable	Scope	Type	Description
iShifter	INPUT	INT	Amount to shift by
iIdx	INPUT	INT	Current index
iMax	INPUT	INT	Maximum index

Example:

```
iShiftedCount := F_SHFT(iShifter, iCount, 2);
```

With an array of 3 pumps, iShifter values of 0, 1, 2 make iShiftedCount start the cascade at index 0, 1, or 2 respectively.

F_NEG

Negates an INT value.

Variable	Scope	Type	Description
iIn	INPUT	INT	Input value to negate

Example:

```
a := F_NEG(120) (* A = -120 *)
b := F_NEG(-1254) (* A = 1254 *)
iDiff := F_NEG(iSetpoint - iActual);
(* iDiff = -(iSetpoint - iActual) *)
```

> **Note:** F_NEG returns INT; set the function return type to INT in the GX Works 2 POU properties.

> `-32768` has no positive counterpart in the `INT` range, so `F_NEG(-32768)` returns `-32768`. Use `F_DNEG` when that value must be handled correctly.

F_DNEG

The 32-bit counterpart of `F_NEG`.

Variable	Scope	Type	Description
`diIn`	INPUT	DINT	Input value to negate

> **Note:** `F_DNEG` returns `DINT`; set the function return type to `DINT` in the GX Works 2 POU properties.
>
> `-2147483648` has no positive counterpart in the `DINT` range, so `F_DNEG(-2147483648)` returns `-2147483648`.

Progress and Time Functions

F_WORK_LEFT

Computes the remaining progress of a process as a backward countdown from 100 to 0, given the elapsed and total time in abstract units (seconds, 100 ms ticks, etc.).

Variable	Scope	Type	Description
`iEt`	INPUT	INT	Elapsed time
`iTw`	INPUT	INT	Total time

Example:

```
OUT_T(xStart, TC0, 200);
iTimeLeft := F_WORK_LEFT(TN0, 200);
```

F_WORK_LEFT_TIME

TIME version of `F_WORK_LEFT`.

Variable	Scope	Type	Description
`tEt`	INPUT	TIME	Elapsed time
`tTw`	INPUT	TIME	Total time

Example:

```
fbTON(IN := xStart, PT := T#5m);
iTimeLeft := F_WORK_LEFT_TIME(fbTON.ET, fbTON.PT);
```

F_VALVE_POS

Returns the current valve position as a percentage `0..1000` (`0.0..100.0 %`) from the elapsed travel time and the total travel time.

Variable	Scope	Type	Description
`tCt`	INPUT	TIME	Current (elapsed) time
`tTt`	INPUT	TIME	Total travel time

Example:

```
iPosition := F_VALVE_POS(tElapsed, tTotal);
```

Other Functions

F_FLT10

Divides an `INT` by 10 and returns the result as a `REAL`.

Variable	Scope	Type	Description
`iIn`	INPUT	INT	Input value

Example:

```
rResult := F_FLT10(123);   (* 12.3 *)
```

> **Note:** `F_FLT10` returns `REAL`; set the function return type to `REAL` in the GX Works 2 POU properties.

L02_SET_IP

Writes an L02 PLC IP setting (PLC IP, gateway, mask, or remote EIP coupler) into the corresponding R register while `xInit` is `TRUE`, and sets `M8197` so the PLC applies the new addresses and restarts the interface. `M8197` resets itself inside the PLC, so neither the application nor the function clears it.

Variable	Scope	Type	Description
`xInit`	INPUT	BOOL	Trigger. The setting is written while this input is `TRUE`
`iWts`	INPUT	INT	Setting type — one of the `IP_*` constants
`wIp1`	INPUT	WORD	IP octet 1
`wIp2`	INPUT	WORD	IP octet 2
`wIp3`	INPUT	WORD	IP octet 3
`wIp4`	INPUT	WORD	IP octet 4

Setting types:

- `IP_PLC_IP` — IP address of the PLC
- `IP_PLC_GATEWAY` — PLC gateway
- `IP_PLC_MASK` — PLC subnet mask
- `IP_REMOTE1` … `IP_REMOTE4` — remote EIP coupler addresses

Example:

```
(* M8002 = first scan, xSaveIp = HMI "save" pulse. *)
xApplyIp := M8002 OR MEP(xSaveIp);

(* All settings of one configuration share the same one-scan trigger. *)
xDone := L02_SET_IP(xApplyIp, IP_PLC_IP, 192, 168, 1, 100);
xDone := L02_SET_IP(xApplyIp, IP_PLC_GATEWAY, 192, 168, 1, 1);
xDone := L02_SET_IP(xApplyIp, IP_PLC_MASK, 255, 255, 255, 0);
```

> **Note:** `L02_SET_IP` performs no return assignment, so its result is always the default (`FALSE`). It is used for its device side effect (writing the R registers).

> **Applying the addresses:** every write ends with `M8197 := TRUE`, which makes the PLC apply the addresses and restart the interface. `M8197` clears itself, so no application code is needed — do not write `M8197` in the user program.

> **Trigger (`xInit`):** the setting is written while this input is `TRUE`, so pass a bit that is `TRUE` for **one scan only** (`M8002` for the startup write, `MEP` of an HMI "save" pulse for a later change) and never tie `xInit` to `TRUE` — the interface would be restarted on every scan. Several settings may be written in the same scan with the same trigger; the last write applies them all.

F_MBMOV

Block-moves `iNum` words from a source device to a destination device while `xEnable` is `TRUE`. The source and destination are given by device number (e.g. `100` = `D100`).

Variable	Scope	Type	Description
`xEnable`	INPUT	BOOL	Enable the move
`iSrc`	INPUT	INT	Source device number
`iNum`	INPUT	INT	Number of words to move
`iDst`	INPUT	INT	Destination device number

Example:

```
xDone := F_MBMOV(TRUE, 100, 4, 200);
(* Copies D100..D103 to D200..D203 *)
```

> **Note:** `F_MBMOV` performs no return assignment, so its result is always the default (`FALSE`). It is used for its device side effect.

Regulators

FB_HYST

On/off regulator function block for heating logic. It turns the output off when `iPV` drops below `iSV`, and on when `iPV` rises above `iSV + iDV`.

Variable	Scope	Type	Description
`xIn`	INPUT	BOOL	Enable regulator
`iSV`	INPUT	INT	Set value
`iPV`	INPUT	INT	Process value
`iDV`	INPUT	INT	Delta (hysteresis band)
`xQ`	OUTPUT	BOOL	Regulator output

Example:

```
VAR
    fbHYST : FB_HYST;
END_VAR

fbHYST(
    xIn := xStart,
    iSV := 255,           (* Set value is 25.5 *)
    iPV := AI_Temperature,
    iDV := 2,             (* Delta is 0.2 *)
    xQ := Y0
);
```

FB_HYST_COOL

Cooling version of `FB_HYST` — the logic is reversed: it turns the output off when `iPV` rises above `iSV`.

Variable	Scope	Type	Description
`xIn`	INPUT	BOOL	Enable regulator
`iSV`	INPUT	INT	Set value
`iPV`	INPUT	INT	Process value
`iDV`	INPUT	INT	Delta (hysteresis band)
`xQ`	OUTPUT	BOOL	Regulator output

Example:

```
VAR
    fbHystCool : FB_HYST_COOL;
END_VAR

fbHystCool(xIn := xStart, iSV := 250, iPV := iTemperature, iDV := 5, xQ := Y0);
```

Analog Input Scaling

Sensor Types

The `STYPE_*` constants select the sensor type for `FB_SCALE_AI` and `L02_SCALE_AI`.

Scalable — these three types are scaled into engineering units using `iMin`/`iMax`:

- `STYPE_0_10V`
- `STYPE_0_20MA`
- `STYPE_4_20MA`

Non-scalable — these types are returned as-is (`iMin`/`iMax` are not applied), but it is still recommended to read them through the block so the internal system devices that configure the AI type are set correctly:

- `STYPE_PT100`
- `STYPE_PT1000`
- `STYPE_TC_K`
- `STYPE_NTC`
- `STYPE_NTC10K`
- `STYPE_TC_E`
- `STYPE_TC_T`
- `STYPE_TC_S`
- `STYPE_TC_J`

`FB_SCALE_AI`

Scales an analog input of the host PLC into engineering units. Coolmay PLCs store analog inputs in `D8030`–`D8045`; the input is 12-bit (0..4000).

Variable	Scope	Type	Description
`iAINum`	INPUT	INT	Analog input number, 0..16
`iSType`	INPUT	INT	Sensor type — see the list above
`iMin`	INPUT	INT	Minimum of the measured unit (scalable types only)
`iMax`	INPUT	INT	Maximum of the measured unit (scalable types only)
`iFilterTime`	INPUT	INT	Filter time, 1..60 ms
`iFilterNum`	INPUT	INT	Filter cycles, 1..999 (default 100)
`iCorrection`	INPUT	INT	Correction added to the output

Variable	Scope	Type	Description
iValueOut	OUTPUT	INT	Scaled value
xErrWire	OUTPUT	BOOL	Wire error
xErrLimit	OUTPUT	BOOL	Input values error (minimum greater than maximum)
xErrMinMax	OUTPUT	BOOL	Min/Max invalid error

Example:

```
VAR
    fbScale : FB_SCALE_AI;
    iPressure : INT;
    iTemperature : INT;
END_VAR

fbScale(
    iAINum := 0,
    iSType := STYPE_4_20MA,
    iMin := 0,
    iMax := 160,           (* 160 is equal to 16.0 bar *)
    iFilterTime := 30,
    iFilterNum := 200,
    iValueOut := iPressure
);

IF fbScale.xErrWire THEN
    (* Wire problem on analog input AD0 *)
END_IF;

fbScale(
    iAINum := 1,
    iSType := STYPE_PT100,
    iCorrection := 5,      (* Add +0.5 *)
    iValueOut := iTemperature
);
```

The same FB instance can read all analog inputs — a new instance is not required for each AI.

L02_SCALE_AI

Scales an analog input of an L02 series module (L02-4AD, L02-RTD, ...) into engineering units. Values are stored in R23700–R23749, 16-bit, range 0..32000.

Variable	Scope	Type	Description
iAINum	INPUT	INT	Analog input number, 0..49
iSType	INPUT	INT	Sensor type — see the list above
iMin	INPUT	INT	Minimum of the measured unit (scalable types only)

Variable	Scope	Type	Description
iMax	INPUT	INT	Maximum of the measured unit (scalable types only)
iCorrection	INPUT	INT	Correction added to the output
iValueOut	OUTPUT	INT	Scaled value

Example:

```
VAR
    fbScaleL02 : L02_SCALE_AI;
    iPressure : INT;
    iTemperature : INT;
END_VAR

fbScaleL02(
    iAINum := 0,
    iSType := STYPE_4_20MA,
    iMin := 0,
    iMax := 160,
    iValueOut := iPressure
);

fbScaleL02(
    iAINum := 1,
    iSType := STYPE_PT100,
    iCorrection := -2,      (* Subtract 0.2 *)
    iValueOut := iTemperature
);
```

Valve Control

FB_VALVE_3P

Controls a 3-position valve. It is not a pulse regulator but a regulator with constant position search: it tracks the current position via the travel time and drives the open/close outputs toward the set position iSV (0..1000).

> **Important:** This function block requires the **TimeControl** library and its TCO_DINT_50 ticker setup.

Variable	Scope	Type	Description
xEnable	INPUT	BOOL	Enable valve control
iSV	INPUT	INT	Set valve position, 0..1000 (0.0..100.0 %)
iDlt	INPUT	INT	Dead zone: no movement while the error is below this value
tTotalTime	INPUT	TIME	Total travel time from fully closed to fully open (add ~2 %)
tLuftTime	INPUT	TIME	Backlash (luft) compensation time on direction change

Variable	Scope	Type	Description
xCloseOnDisable	INPUT	BOOL	Close the valve when control is disabled
xIsOpened	INPUT	BOOL	Fully-open feedback
xIsClosed	INPUT	BOOL	Fully-closed feedback
xOpen	OUTPUT	BOOL	Open-valve command
xClose	OUTPUT	BOOL	Close-valve command

Example:

```
VAR
    fbValve : FB_VALVE_3P;
END_VAR

fbValve(
    xEnable := X0,
    iSV := iPidTask,
    iDlt := 50,                 (* 5.0 % *)
    tTotalTime := T#10s,
    tLuftTime := T#500ms,
    xCloseOnDisable := TRUE,
    xIsOpened := X2,
    xIsClosed := X3,
    xOpen := Y0,
    xClose := Y1
);
```